

Project 4: Preference Elicitation

User Case Study

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Abstract

The user case study is about a system that learns a user's movie preferences from a few comparisons. We describe how movies from the IMDB 5000 dataset are represented as feature vectors, how the Bradley-Terry model is extended from pairwise comparisons to full rankings and how a user study comparing a pairwise interface with a ranking interface would be designed.

1 Introduction

Used dataset: IMDB 5000 Movie Dataset

Source: <https://www.kaggle.com/datasets/carolzhangdc/imdb-5000-movie-dataset>

Each movie i is described by a feature vector x_i . The user's taste is described by a weight vector w with one weight per feature. The utility of a movie for the user is $u_i = w^\top x_i$ and the system recommends the movies with the highest utility. This is a content-based recommender [2] as introduced in the 8th lecture. The main challenge is that w must be learned from only a few answers i.e. movie selection of a single user.

2 Feature Representation

2.1 Data preparation

The dataset contains 5043 movies with 28 attributes and contains invisible characters or leading spaces which we removed. Many values are missing, especially in the features box-office gross (17.5%) and the budget (9.8%). Deleting these movies would remove a big portion of the dataset, mostly smaller productions. Instead, missing numeric values are replaced by the median and a flag marks that the value was missing. Missing categorical values get their own category *Unknown*.

2.2 Features

Table 1 lists the features that are used while learning. Because every feature adds one weight that must be learned from few answers, we only keep features that influence user's movie preferences and occur often enough to be learnable.

- **Genres and keywords** are multi-hot-encoded because a movie can have several at once. Only the 30 most frequent keywords are kept: the dataset has 8086 different keywords, most of which occur in a single movie and could never be learned.
- **Country and content rating** keep their five most frequent values. Rare values are grouped into *Other*.

Group	Columns	Encoding
Color	3	one-hot (Color, Black and White, Unknown)
Numeric attributes	14	median-imputed, standardized
Missing indicators	14	1 if the value was missing
Gross/Budget ratio	1	derived, standardized
Genres	26	multi-hot-encoded
Content rating	7	top 5 + Other + Unknown
Country	7	top 5 + Other + Unknown
Plot keywords	30	multi-hot, 30 most frequent keywords
Total	102	

Table 1: Feature representation of a movie

- **Money and count attributes** (budget, gross, votes, reviews, Facebook likes) are log-transformed, because a few blockbusters have extremely large values (e.g. budgets of several hundred million dollars). All numeric features are then standardized so that all weights are on a comparable scale.
- **Gross/Budget ratio**: a linear model can already combine gross and budget additively, so a feature like profit would add nothing new. A ratio cannot be formed this way, so it is added as a measure of commercial success relative to cost (e.g. a low-budget movie that became a big commercial success). It is computed only where both values are known and capped at its 99th percentile to limit extreme outliers.
- **Excluded features**: actor and director names (thousands of different values), language (93.5% English), aspect ratio and number of faces on the poster (no plausible relation to user’s movie preferences).

3 Extending Bradley-Terry to Rankings

3.1 Bradley-Terry model

For a pairwise comparison, the Bradley-Terry model [5, 4] gives the probability that the user prefers movie i over movie j :

$$P(i \succ j) = \frac{e^{u_i}}{e^{u_i} + e^{u_j}}. \quad (1)$$

3.2 Proposed extension

For a ranking $\pi = (\pi_1, \dots, \pi_k)$ of k movies, where π_1 is the favorite, we use the Plackett-Luce model [8]:

$$P(\pi) = \prod_{t=1}^{k-1} \frac{e^{u_{\pi_t}}}{\sum_{s=t}^k e^{u_{\pi_s}}}. \quad (2)$$

The ranking is seen as a sequence of choices: the user first picks the favorite out of all k movies, then the favorite out of the remaining ones and so on. Each step is a Luce choice [7, 3], thus an option is chosen with a probability proportional to e^u .

3.3 Justification

- **It generalizes Bradley-Terry.** For $k = 2$, equation (2) is exactly the Bradley-Terry model. Both interfaces estimate same weights w and a pairwise answer is simply a ranking of length two.

- **It matches how people rank.** Ordering a list step by step, from the best to the worst remaining option, is a natural way to produce a ranking.
- **It avoids a common mistake.** Splitting a ranking into all its pairs and treating them as independent comparisons would overcount the evidence because these pairs come from one single decision.

3.4 Learning the user's weights

With 102 weights and only a few answers, many different weight vectors explain the answers equally well. We therefore add a penalty that prefers small weights which corresponds to a Gaussian prior on w [3] and minimize

$$\mathcal{L}(w) = - \sum_n \log P(\pi^{(n)}) + \frac{\lambda}{2} \|w\|^2, \quad \lambda = 1. \quad (3)$$

The minimum is found numerically. In addition, the curvature of $\mathcal{L}$ at the minimum gives an estimate of how certain the model is about w (Laplace approximation) that is shown to the user after the study.

4 User Study Design

The design follows the content of lecture 7 [1].

4.1 Research question and hypotheses

Which interface learns a user's preferences more efficiently? Pairwise comparisons or rankings of 10 movies? A ranking of 10 movies contains much more information than a single pairwise choice but it also takes more effort. This trade-off depends on human behavior and can only be answered with a user study.

- **Hypothesis 1:** The ranking interface predicts the user's preferences more accurately.
- **Hypothesis 2:** The ranking interface takes more time and effort per movie shown.

4.2 Design

We use a **between-subjects** design. Each participant uses only one interface, assigned at random by the system. A within-subjects design would need fewer participants but the second interface would be influenced by the first one, e.g. because participants have already thought about their taste. Random assignment ensures that differences between the groups are caused by the interface and not by the participants choosing their preferred interface.

The two conditions show a similar number of movies: 20 pairwise questions (i.e. 40 movies) or 5 rankings of 10 movies (i.e. 50 movies). In both conditions, the movies are drawn randomly from the dataset.

4.3 Measurements

- **Accuracy:** at the end, every participant answers the same 20 pairwise test questions, which are not used for learning. Accuracy is the share of test questions for which the model predicts the participant's choice correctly.
- **Time:** total time and time per movie, taken from the logged timestamps.
- **Perceived effort and satisfaction:** two short questions on a scale from 1 to 5.

4.4 Participants and recruitment

Participants are adults (18+) who watch movies at least occasionally and understand English, the language of the interface. They are recruited through university mailing lists/AStA-newsletter and student forums. A standard power analysis (medium effect $d = 0.5$, $\alpha = 0.05$, power 0.8) [6] gives 64 participants per group. We plan 75 per group to allow for dropouts.

4.5 Procedure

1. Information about the study and informed consent
2. A short survey (age, how often they watch movies)
3. Instructions and one practice question
4. Main phase with the assigned interface
5. The 20 test questions
6. The questions on effort and satisfaction
7. Debriefing and the list of recommendations

Before the main study, a pilot with 5 to 10 participants checks the duration and the clarity of the instructions. Their data are not used [1].

4.6 Ethics and data protection

Participants give informed consent and can stop the study at any time. Only their answers and the timestamps of each question are stored under a random number and without names or other personal data. Participants can request the deletion of their data, in line with the GDPR [1].

4.7 Analysis

Incomplete sessions and sessions with implausibly fast answers (clicking blindly) are excluded, using criteria fixed before data collection [1]. The two groups are compared with a t -test at $\alpha = 0.05$. Since two hypotheses are tested, a Bonferroni correction is applied.

5 Implementation

The study interface is implemented as a Django application. After giving consent, each participant is randomly assigned to one interface. Rankings are entered by clicking the movies in order of preference. A pairwise question works the same way with two movies, i.e. one page serves both interfaces. All answers are stored in a database together with the time each question was shown and answered. At the end, the model from Section 3 is fitted and the ten best unseen movies are recommended together with the model’s certainty. The test questions and questionnaire from Section 4 would be added before running the study.

References

- [1] Human-Centric Artificial Intelligence, Lecture 7: User Studies. TUHH, Summer Term 2026.
- [2] Human-Centric Artificial Intelligence, Lecture 8: Interactive ML. TUHH, Summer Term 2026.

- [3] Human-Centric Artificial Intelligence, Lecture 9: User Modelling. TUHH, Summer Term 2026.
- [4] Human-Centric Artificial Intelligence, Lecture 11: AI Safety. TUHH, Summer Term 2026.
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